Sarvan Sir's Economy Notes

NATIONAL INCOME ACCOUNTING

GDP Product Method (Value Added Method / Output Method)

Gross Domestic Product (GDP) is the total value of all final goods and services produced within the domestic territory of a country by resident units during a specific period of time.

1. Resident Unit

An individual or institutional unit having a predominant economic interest within the economic territory of a country and generally performing economic activities for more than one year.

Examples:

- A foreign employee staying and working in India for two years is generally considered a resident unit of India.
- A foreign tourist visiting India for two months is considered a non-resident unit.
- An Indian citizen permanently settled in the USA is considered a non-resident unit of India.

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2. Domestic Economic Territory

Includes:

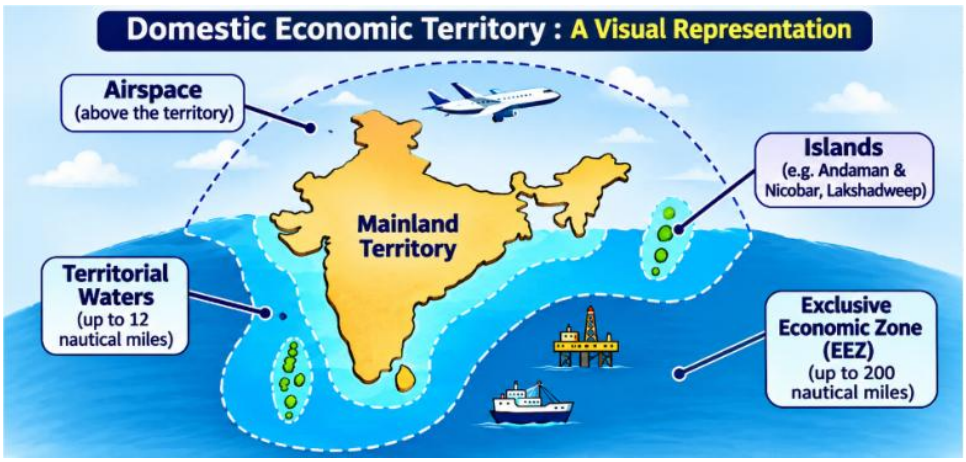
- Mainland territory of the country.
- Islands belonging to the country.
- Territorial waters.
- Airspace above the domestic territory.
- Exclusive Economic Zone (EEZ), to the extent that the country exercises exclusive economic rights over resources and activities.
- Indian embassies, consulates and military establishments located in foreign countries.

Does not include:

- Foreign embassies and consulates located within India.
- Foreign military establishments located within India.

Note: Economic territory is not exactly the same as geographical territory. Indian embassies abroad are treated as part of India's economic territory, whereas foreign embassies in India are treated as part of the respective foreign country's economic territory.

Domestic Economic Territory Visual Representation



- **Includes (Domestic Economic Territory):**
  - Mainland territory
  - Islands (e.g., Andaman & Nicobar, Lakshadweep)
  - Territorial waters (up to 12 nautical miles)
  - Airspace (above the territory)
  - Exclusive Economic Zone (EEZ) (up to 200 nautical miles)
  - Indian embassies, consulates and military establishments abroad
- **Does Not Include:**
  - Foreign embassies and consulates within India
  - Foreign military establishments within India

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Product Method / Value Added Method / Output Method

1. Meaning / Definition

Under this method, GDP is calculated by aggregating the value added at each stage of production (or the value of final goods and services) to arrive at the total value of output produced in the economy.

2. Formula (GDP at Market Price)

$$\text{GDP (at Market Price)} = \text{GVA (at Basic Prices)} + \text{Net Product Taxes}$$

3. GVA at Basic Prices - Formula (MOSPI)

$$\text{GVA (at Basic Prices)} = \text{CE} + \text{OS/MI} + \text{CFC} + \text{Taxes on Production} - \text{Subsidies on Production}$$

4. Components of GVA (Factor Payments)

- **CE (Compensation to Employees):** Salaries, wages and employers' social contributions.
- **OS (Operating Surplus):** Income of incorporated enterprises from production (profit, rent and interest).
- **MI (Mixed Income):** Income of self-employed persons and owners of unincorporated enterprises (combining labour and capital).
- **CFC (Consumption of Fixed Capital):** Depreciation on fixed assets due to wear and tear, ageing and obsolescence.
- **Taxes on Production:** Taxes levied on production by the government, such as property tax, business licence fees, payroll tax, etc.
- **Subsidies on Production:** Financial assistance provided by the government to producers towards production-related costs and employment.

5. Meaning of GVA at Basic Prices

Gross Value Added (GVA) at Basic Prices is the value of output produced by an economic unit, measured at basic prices (producer's price), minus the value of intermediate consumption.

- **Basic Price (Producer's Price):**
  - The amount receivable by the producer from the purchaser for a unit of output, excluding taxes on products and including subsidies on products.

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6. Alternative Formula

$$\text{GVA at Basic Prices} = \text{Output} - \text{Intermediate Consumption}$$

Important Note - Basic Price vs Market Price

- ***Basic Price:*** *Excludes taxes on products (e.g., GST, excise duty and customs duty) and includes subsidies on products.*
- ***Market Price:*** *Includes taxes on products and excludes subsidies on products.*

Illustration: GDP Calculation Farmer and Bakery (Product Method)

Example: Farmer - Bakery - Consumer

Let us understand the calculation of GDP and product tax with a simple example of a farmer and a bakery.

Step 1: Farmer - Production and Sale of Wheat

- Total wheat produced = ₹1,000
- Retained for own consumption = ₹600

- Sold to bakery owner = ₹400
- GST @ 10% on ₹400 = ₹40
- Farmer receives ₹440 (₹400 + ₹40)
- Farmer remits GST of ₹40 to Government.



Step 2: Bakery - Value Addition and Output

- Input (Wheat) purchased = ₹440 (₹400 + ₹40 GST @ 10%)
- Value added by the bakery = ₹500
- Value of bread (before GST) = ₹400 + ₹500 = ₹900
- Add GST @ 10% on ₹900 = ₹90
- Final price to consumer = ₹990

The ₹40 (GST paid on input) is an Input Tax Credit (ITC) and not included again while computing output tax.



Step 3: GST Settlement, Government Receipt and GDP Calculation

- **(A) GST Settlement by Bakery**
  - Output tax collected (on bread) = ₹90
  - Input tax already paid (on wheat) = ₹40
  - Net GST payable to Government (₹90 - ₹40) = ₹50

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- **(B) Total Product Tax Received by Government**
    - From Farmer = ₹40
    - From Bakery (Net) = ₹50
    - **Total Tax Received = ₹90** *(Government receives the tax only on the final goods = ₹90)*.
- **(C) Calculation of GDP**

Final Goods	**Value(₹)**
Wheat (own consumption)	600
Bread (sold to ultimate consumer)	990
Total GDP	**1590**

➢ **GDP:** The final value of all goods and services (monetary value) produced in the economy.

Conclusion: GDP of this economy = ₹1,590 and Product tax received by the Government = ₹90.

Key Takeaway - Product Method

The value added at different stages of production, when aggregated, gives the total value of final goods and services produced in the economy. This method avoids double counting of intermediate goods.

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PRODUCT METHOD - CONCLUDING POINTS

1. DIFFERENCE BETWEEN GVA AND GDP

- **GVA (Gross Value Added):** GVA measures the value added by a producer or production unit during the process of production. It is calculated as the value of output minus intermediate consumption.
- **GDP (Gross Domestic Product):** GDP measures the total monetary value of all final goods and services produced within the domestic territory of a country during a specific period.

2. CHANGE IN INDIA'S NATIONAL ACCOUNTS - 2015

Adoption of the 2008 SNA

In 2015, India introduced major changes to its National Accounts compilation methodology and aligned it with the United Nations' System of National Accounts 2008 (2008 SNA).

SNA = System of National Accounts